



Organic Chemistry: Techniques & Spectra

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Reflux, Distillation & Boiling Point Determination

Heating under reflux

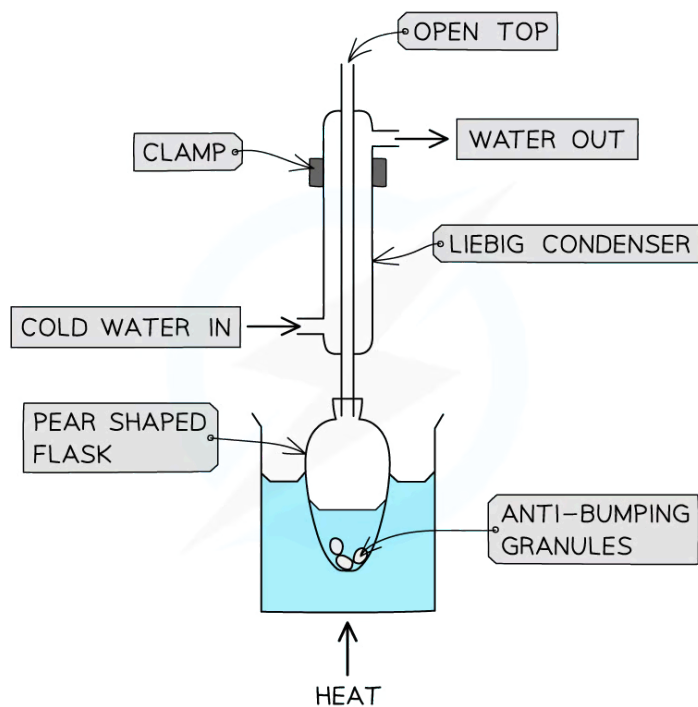
- Organic reactions often occur slowly at room temperature
- Therefore, organic reactions can be completed by heating under reflux to produce an **organic liquid**
- This allows the mixture to react as fully as possible without the loss of any reactants, products or solvent
 - In distillation, you are trying to separate a chemical or product from a mixture
 - When heating under reflux, you aim to keep all the chemicals inside the reaction vessel

The Heating under Reflux Process:

- Example reactions where heating under reflux could be used include:
 - The production of a carboxylic acid from a primary alcohol using acidified potassium dichromate
 - The production of an ester from an alcohol and acid in the presence of an acid catalyst
- The reaction mixture is placed into a pear-shaped or round bottomed flask
- Anti-bumping granules are, again, added to promote smooth boiling
- The flask is placed in a heating mantle or it can be immersed in a water bath for heating
- Quickfit apparatus is then set up with the condenser clamped vertically in place
 - The joints of the Quickfit apparatus are commonly greased as with distillation
- A steady and constant stream of water passes through the condenser in a 'water jacket' - it enters at the bottom of the condenser and the drainage pipe removes the water from the top of the condenser
- The water is heated and the reaction mixture allowed to boil
- The heated is stopped and the mixture allowed to cool back to room temperature



Your notes



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The preparation of ethyl ethanoate involves heating under reflux for about 15 minutes

Distillation

- Distillation is a common practical completed in organic chemistry
- Distillation is used as there are times that a reaction does not go to completion or there are other chemicals produced as well as the desired product
- Distillation allows you to separate compounds by their boiling point
 - Chemicals with the lowest boiling point will distill first
- One of the most common distillation practicals is the oxidation of primary and secondary alcohol to aldehydes and ketones

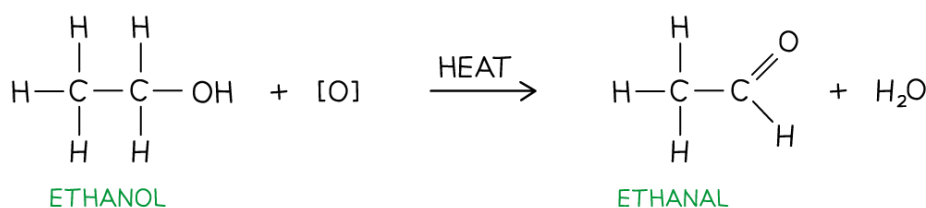
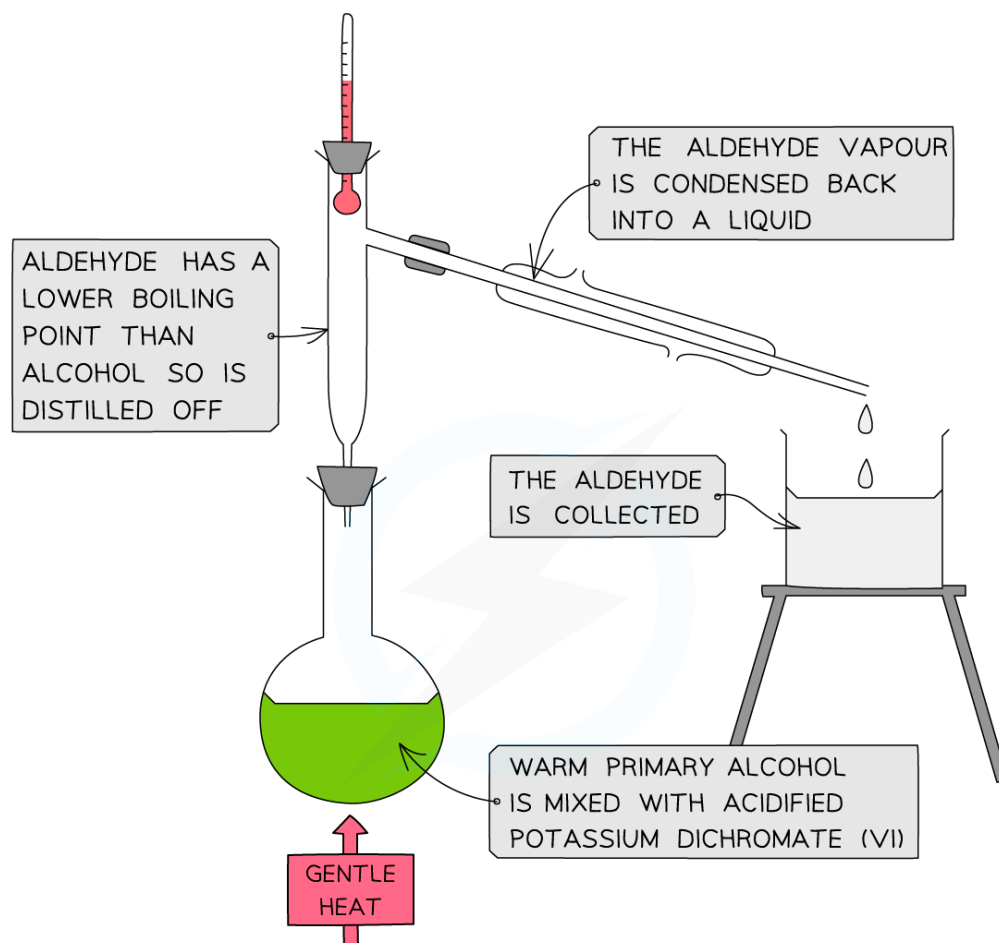
The Distillation Process:

- To produce an aldehyde from a primary alcohol, a reaction mixture containing the primary alcohol and acidified potassium dichromate solution is placed into a pear-shaped or round bottomed flask
- Anti-bumping granules are added to promote smooth boiling
- Quickfit apparatus is then set up, including a still head and condenser connected to the side
 - The joints of the Quickfit apparatus are often have a thin layer of silicon grease smeared over them to give a better seal as well as to make it easier to disassemble the equipment afterwards

- A Quickfit thermometer can be used, with the thermometer bulb sitting where the vapours will pass into the condenser
- A steady and constant stream of water passes through the condenser in a 'water jacket' - it enters at the bottom of the condenser and the drainage pipe removes the water from the top of the condenser



Your notes



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Heating under Distillation Apparatus

- The reaction mixture is heated until it boils using a heating mantle
 - Electric heating mantles are used for this because the temperature can be controlled, and because you are using chemicals which are flammable
- The distillate which forms in the condenser drips directly into a receiving vessel
 - The distillate which should be collected, is that which is given off at $\pm 2^\circ\text{C}$ of the boiling point of the desired product



- Some distillate may be given off below this temperature - this needs to be discarded and a clean vessel used to collect the desired product
- Stop collecting the distillate if the temperature rises above $\pm 2^{\circ}\text{C}$ of the boiling point of the desired product
- The **aldehyde** product has a lower boiling point than the **alcohol** (since it has lost the **H-bonding**) so it can be **distilled off** as soon as it forms



Examiner Tips and Tricks

- These practicals give you the opportunity to discuss:
 - The use of an electric heating mantles and water baths rather than a Bunsen burner
 - The choice and setup of laboratory apparatus
 - Health and safety considerations including the careful handling of different liquids, including those which are corrosive, irritant, flammable and toxic

Boiling point determination

- The boiling point of a liquid is indicative of its purity and identity
- Boiling point is determined by **distillation**
- The sample is gently heated until it boils and this temperature is recorded
 - This boiling point can then be compared against literature / database values
- If the sample contains impurities:
 - The boiling point may appear higher than the literature / database values
 - The sample may boil over a range of temperatures instead of at a single temperature

Solvent Extraction & Drying

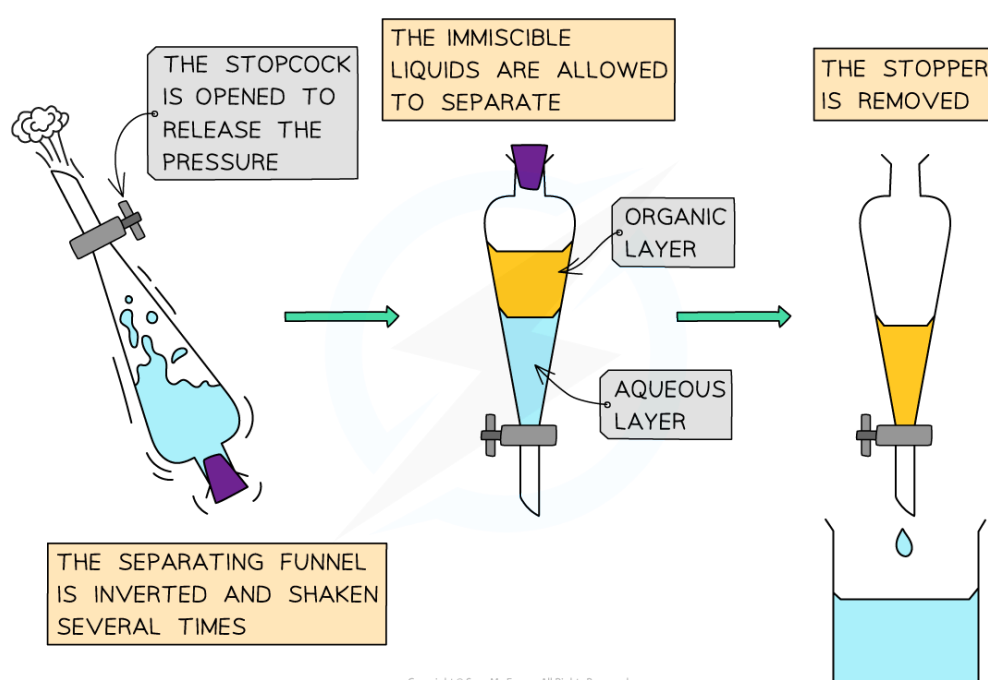
- There are different methods to purify organic compounds, including:
 - Solvent extraction / use of a separating funnel
 - Use of drying agents

Solvent extraction / use of a Separating Funnel

- When organic liquids are being prepared, water can often be obtained along with the organic product
- The water will usually form an aqueous layer with the product in the organic layer
 - It can sometimes be hard to identify which layer is the organic layer - this can be achieved by simply adding water and seeing which layer increases in volume
- Other organic reactions may need to be neutralised before being purified, this can be achieved by adding sodium carbonate solution to the reaction vessel or separating funnel



- This can also be used to remove other impurities by washing
- The contents of the reaction vessel are transferred to a separating funnel and a stopper added
- The separating funnel is inverted and the stopcock opened to release the pressure - this is repeated 15–20 times
 - If neutralisation has occurred then the stopcock is opened slowly to avoid losing any product
- The two layers are allowed to separate
 - In the following example, the aqueous layer is the bottom layer inside the separating funnel
- The stop cock is opened so that the aqueous layer drains away and the organic layer can be drained into a clean beaker



A separating funnel allows the product to be cleaned and isolated

Use of drying agents

- Drying agents can be used to remove traces of water from an organic product
- Drying agents are usually anhydrous inorganic salts that readily hydrate in the presence of water
 - Anhydrous calcium chloride is commonly used to dry hydrocarbons
 - Anhydrous calcium or magnesium sulfate are used as more general purpose drying agents
- A spatula of drying agent is added into the organic product and swirled



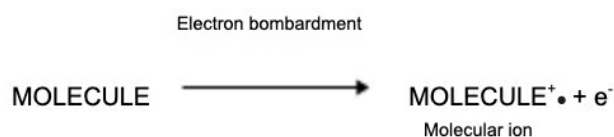
Your notes

- If the organic product has a low boiling point, a lid / stopper can be added to reduce the potential evaporation of any product
- If the drying agent clumps together, then there is still water in the organic liquid
- More drying agent is added until some remains dispersed in the organic liquid as a fine powder
- The dry organic liquid can then be decanted or filtered
 - If the organic liquid is dry then it should appear clear

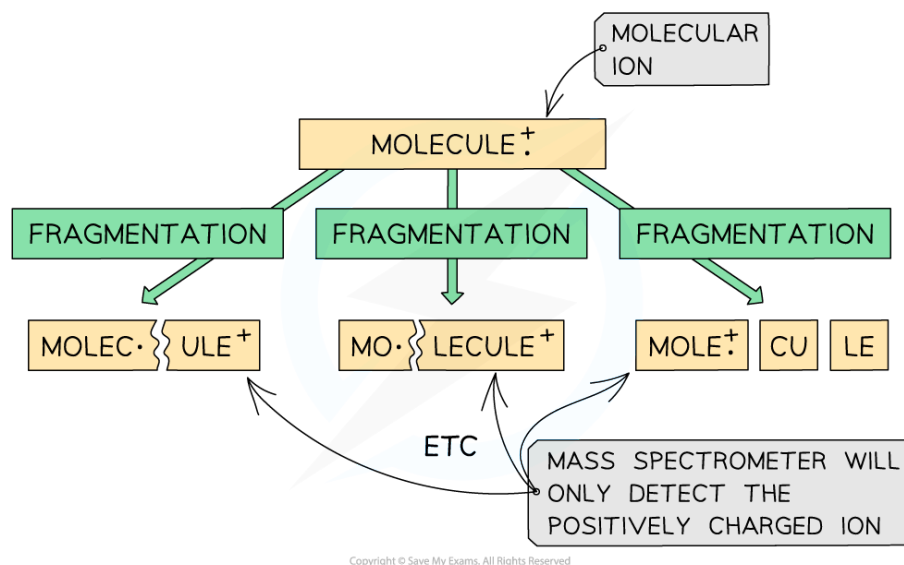


Mass Spectroscopy – Interpretation

- **Mass spectroscopy** is an analytical technique used to identify unknown compounds
- The molecules in the small sample are **bombarded** with high energy electrons which can cause the molecule to lose an electron
- This results in the formation of a positively charged **molecular ion** with one unpaired electron
 - One of the electrons in the pair has been **removed** by the beam of electrons



- The molecular ion can further **fragment** to form new ions, molecules, and radicals



Fragmentation of a molecule in mass spectroscopy

- These **fragmented** ions are **accelerated** by an electric field
- Based on their mass (**m**) to charge (**z**) ratio, the ion fragments are then separated by deflecting them into the **detector**
 - Most ions will only gain a charge of 1+ and therefore a ion with mass 12 and charge 1+ will have an m/z value of 12
 - It is, however, possible for a greater charge to occur. For example, an ion with mass 16 and charge 2+ will have a m/z value of 8

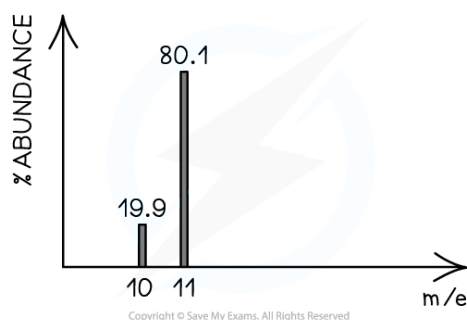


Your notes

- The smaller and more positively charged fragment ions will be **detected** first as they will get **deflected** the most and are more attracted to the **negative pole** of the magnet
- Each fragment corresponds to a specific **peak** with a particular m/z value in the mass spectrum
- The **base peak** is the peak corresponding to the most **abundant** ion
- The m/z is sometimes referred to as the m/e ratio and it is almost always **1:1**

Isotopes

- Isotopes are different atoms of the **same element** that contain the same number of **protons** and **electrons** but a different number of **neutrons**.
 - These are atoms of the same **elements** but with different mass number
 - For example, Cl-35 and Cl-37 are isotopes as they are both atoms of the same element (chlorine, Cl) but have a different mass number (35 and 37 respectively)
- Mass spectroscopy can be used to find the **relative abundance** of the isotopes experimentally
- The **relative abundance** of an isotope is the proportion of one particular isotope in a mixture of isotopes found in nature
 - For example, the relative abundance of Cl-35 and Cl-37 is 75% and 25% respectively
 - This means that in nature, 75% of the chlorine atoms is the Cl-35 isotope and 25% is the Cl-37 isotope
- The **heights** of the peaks in mass spectroscopy show the proportion of each isotope present



The peak heights show the relative abundance of the boron isotopes: boron-10 has a relative abundance of 19.9% and boron-11 has a relative abundance of 80.1%



Worked Example

Calculating m/z ratio

In a sample of iron, the ions $^{54}\text{Fe}^{2+}$ and $^{56}\text{Fe}^{3+}$ are detected. Calculate their m/z ratio and determine which ion is deflected more inside the spectrometer.

Answer

$$m/e (^{54}\text{Fe}^{2+}) = \frac{54}{2}$$

$$= 27$$

$$m/e (^{56}\text{Fe}^{3+}) = \frac{56}{3}$$

$$= 18.7$$

$$= 19$$

- $^{56}\text{Fe}^{3+}$ has a smaller m/z ratio and will therefore be deflected more.
 - It also has the largest positive charge and will be more attracted to the negative pole of the magnet within the mass spectrometer.

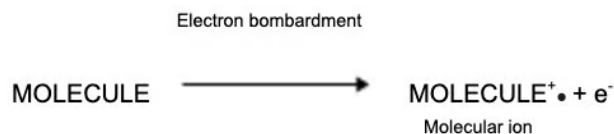


Examiner Tips and Tricks

A small m/z value corresponds to fragments that are either **small** or have a **high positive charge** or a combination of **both**.

Deducing molecular formulae

- Each **peak** in the mass spectrum corresponds to a certain **fragment** with a particular m/z value
- The peak with the highest m/z value is the molecular ion (M^+) peak which gives information about the **molecular mass** of the compound
- The molecular ion is the entire molecule that has lost one electron when bombarded with a beam of electrons



- The **[M+1]** peak is a smaller peak which is due to the natural abundance of the isotope carbon-13
- The amount of naturally occurring C-13 is a little over 1%, so the [M+1] peak is very small
- The height of the [M+1] peak for a particular ion depends on how many carbon atoms are present in that molecule; the more carbon atoms, the larger the [M+1] peak is



Your notes

- For example, the height of the $[M+1]$ peak for an hexane (containing six carbon atoms) ion will be greater than the height of the $[M+1]$ peak of an ethane (containing two carbon atoms) ion



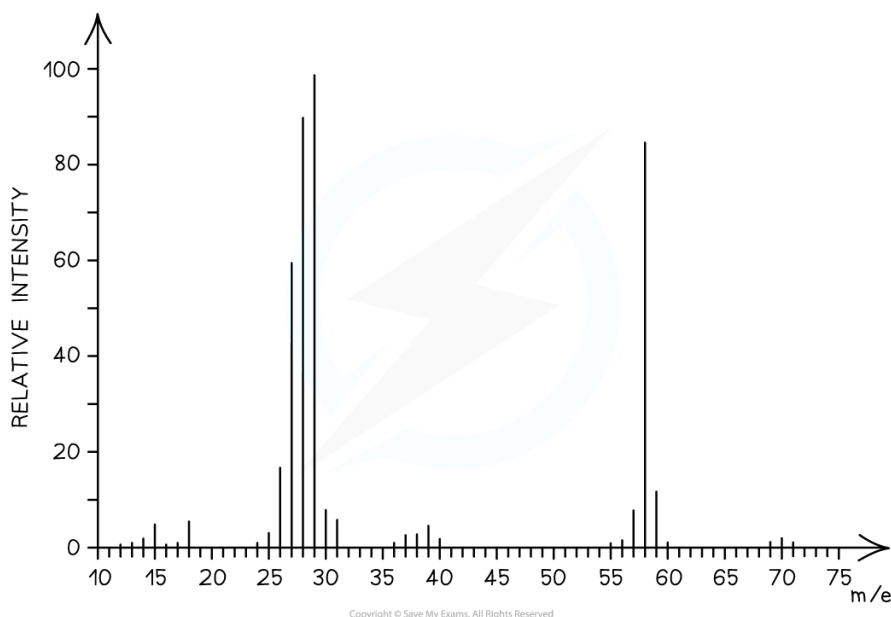
Your notes



Worked Example

Analysing mass spectra

Determine whether the following mass spectrum corresponds to but-1-ene or pent-1-ene:



Answer

- The mass spectrum corresponds to pent-1-ene as the molecular ion peak is at $m/z = 70$
 - The small peak at $m/z = 71$ is a C-13 peak, which does not count as the molecular ion peak
 - But-1-ene arises from the $C_4H_8^+$ ion which has a molecular mass of 56
 - Pent-1-ene arises from the $C_5H_{10}^+$ ion which has a molecular mass of 70

Mass Spectrum – Fragmentation

- The molecular ion peak can be used to identify the **molecular mass** of a compound
- However, different compounds may have the same molecular mass
- To further determine the structure of the unknown compound, **fragmentation** is used
- Fragments may appear due to the formation of **characteristic fragments** or the **loss of small molecules**

- For example, a peak at 29 is due to the characteristic fragment C_2H_5^+
- Loss of small molecules give rise to peaks at 18 (H_2O), 28 (CO), and 44 (CO_2)

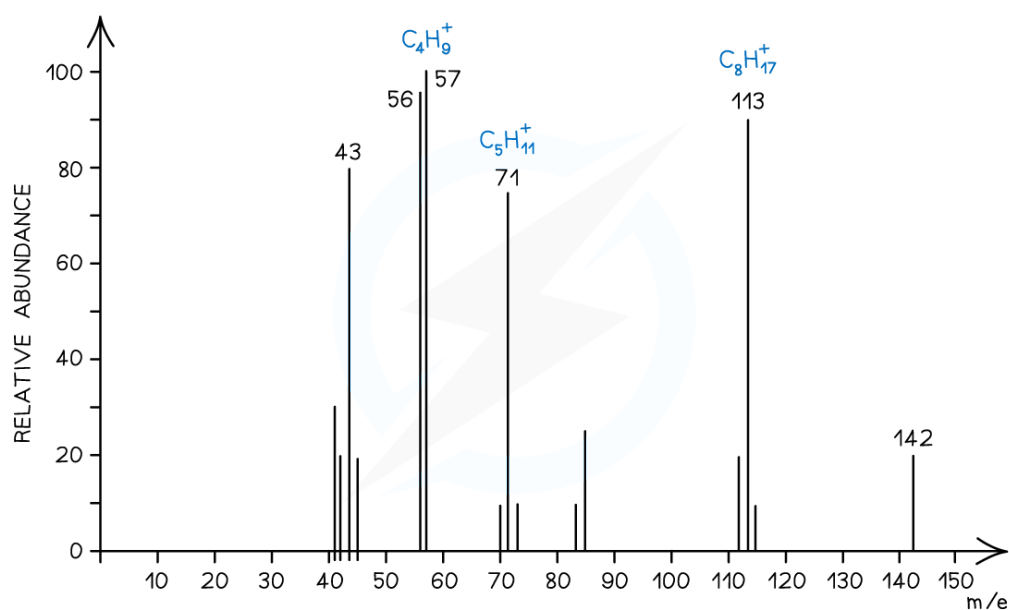
Alkanes

- Simple alkanes are fragmented in mass spectroscopy by breaking the C-C bonds
- m/e values of some of the common alkane fragments are given in the table below

m/e Values of Fragments Table

Fragment	m/e
CH_3^+	15
C_2H_5^+	29
C_3H_7^+	43
C_4H_9^+	57
$\text{C}_5\text{H}_{11}^+$	71
$\text{C}_6\text{H}_{13}^+$	85

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Mass spectrum showing the fragmentation of $\text{C}_{10}\text{H}_{22}$

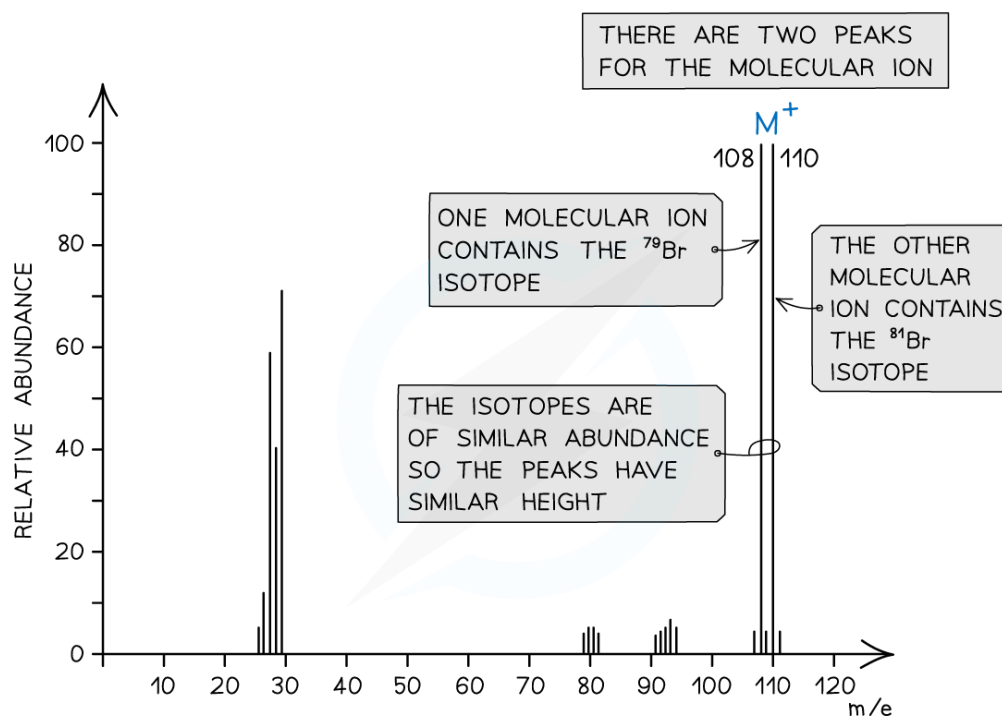
Halogenoalkanes

- Halogenoalkanes often have multiple peaks around the molecular ion peak

- This is caused by the fact that there are different isotopes of the halogens



Your notes



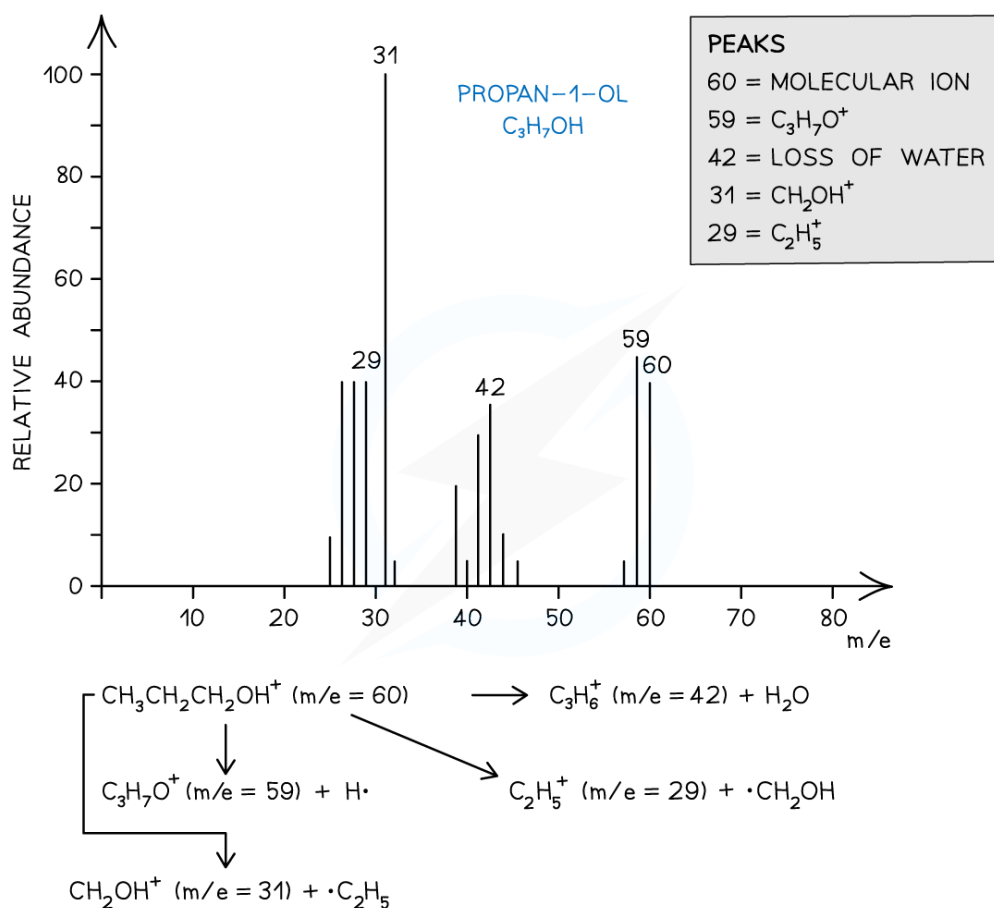
Mass spectrum showing different isotopes of bromine in the molecular ion

Alcohols

- Alcohols often tend to lose a **water molecule** giving rise to a peak at **18 below the molecular ion**
- Another common peak is found at m/e value 31 which corresponds to the CH_2OH^+ -fragment
- For example, the mass spectrum of propan-1-ol shows that the compound has fragmented in four different ways:
 - Loss of $\text{H}^\bullet$ to form a $\text{C}_3\text{H}_7\text{O}^+$ fragment with $m/e = 59$
 - Loss of a water molecule to form a C_3H_6^+ fragment with $m/e = 42$
 - Loss of a $^\bullet\text{C}_2\text{H}_5$ to form a CH_2OH^+ fragment with $m/e = 31$
 - And the loss of $^\bullet\text{CH}_2\text{OH}$ to form a C_2H_5^+ fragment with $m/e = 29$



Your notes



Mass spectrum showing the fragmentation patterns in propan-1-ol (alcohol)



Worked Example

Ion fragmentation

Which of the following statements about the mass spectrum of CH_3Br is correct?

- A. There is one peak for the molecular ion with an m/e value of 44
- B. There is one peak for the molecular ion with an m/e value of 95
- C. The last two peaks have abundances in the ratio 3:1 and occur at m/e values of 94 and 96
- D. The last two peaks are of equal size and occur at m/e values of 94 and 96

Answer

The correct answer is option **D**

- Bromomethane (CH_3Br) can produce 3 peaks

- $\text{CH}_3^{81}\text{Br} \rightarrow [\text{CH}_3^{81}\text{Br}]^+ + \text{e}^-$ at m/e 96
- $\text{CH}_3^{79}\text{Br} \rightarrow [\text{CH}_3^{79}\text{Br}]^+ + \text{e}^-$ at m/e 94
- $\text{CH}_3\text{Br} \rightarrow [\text{CH}_3]^+ + \cdot\text{Br}$ at m/e 15
- The last two peaks (which correspond to the molecular ion peak) therefore are equal in size and occur at m/e values of 94 and 96



Your notes



Worked Example

Alcohol fragmentation

Which alcohol is not likely to have a fragment ion at m/e at 43 in its mass spectrum?

- A. $(\text{CH}_3)_2\text{CHCH}_2\text{OH}$
- B. $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_3$
- C. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
- D. $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$

Answer

The correct answer is option **D**

- Because a line at $m/e = 43$ corresponds to an ion with a mass of 43 for example:
 - $[\text{CH}_3\text{CH}_2\text{CH}_2]^+$
 - $[(\text{CH}_3)_2\text{CH}]^+$
 - 2-butanol is not likely to have a fragment at $m/e = 43$ as it does not have either of these fragments in its structure.



Infrared Spectrometry – Interpretation

- **Infrared (IR) spectroscopy** is a technique used to identify compounds based on changes in **vibrations** of atoms when they **absorb** IR of certain **frequencies**
- A **spectrophotometer** irradiates the sample with electromagnetic waves in the infrared region and then detects the **intensity** of the wavelength of IR radiation which goes through the sample
- All organic molecules absorb IR radiation and depending on which energies of radiation are absorbed, bonds between atoms will vibrate by **stretching**, **bending** and **twisting**
- The molecules will only vibrate at a specific frequency
- The **resonance frequency** is the specific frequency at which the molecules will vibrate to stimulate larger vibrations
- Depending on the rest of the molecule, each vibration will absorb specific wavelengths of IR radiation which are also shown as the **reciprocal** of the wavelength
 - This unit is called the **wavenumber** (cm^{-1})
- Particular absorbance have characteristic **widths** (broad or sharp) and **intensities** (strong or weak)
 - For example, hydrogen bonds cause the O-H bonds in alcohols and carboxylic acids to be **broad** whereas the C-O bond in carbonyl ($\text{C}=\text{O}$) groups have a strong, sharp absorbance peak
- The energies absorbed by different functional groups are given as a range and an unknown compound can be identified by comparing its IR spectrum to the IR spectrum of a known compound

Absorption Range of Bonds Table



Your notes

Bond	Functional groups containing the bond	Characteristic infrared absorption range (in wavenumbers) (cm^{-1})
C-O	Hydroxy, ester	1040–1300
C=C	Aromatic compound, alkene	1500–1680
C=O	Amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	Nitrile	2200–2250
C-H	Alkane	2850–2950
N-H	Amine, amide	3300–3500
O-H	Carboxyl, hydroxyl	2500–3000 3200–3600

- Due to some absorption bands overlapping each other, other analytical techniques such as mass spectroscopy should be used alongside IR spectroscopy to identify an unknown compound



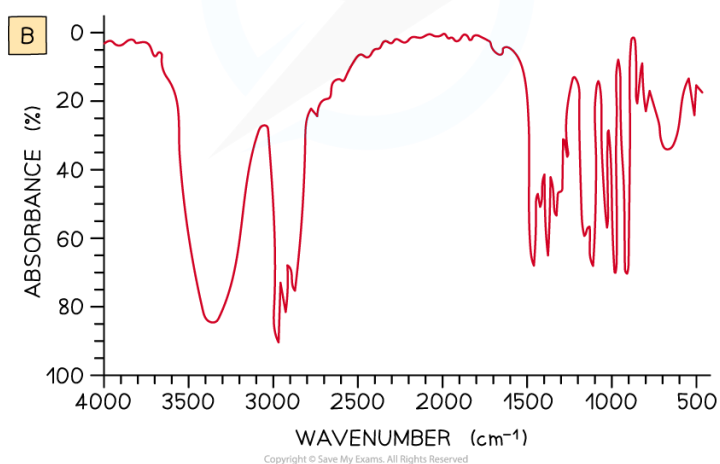
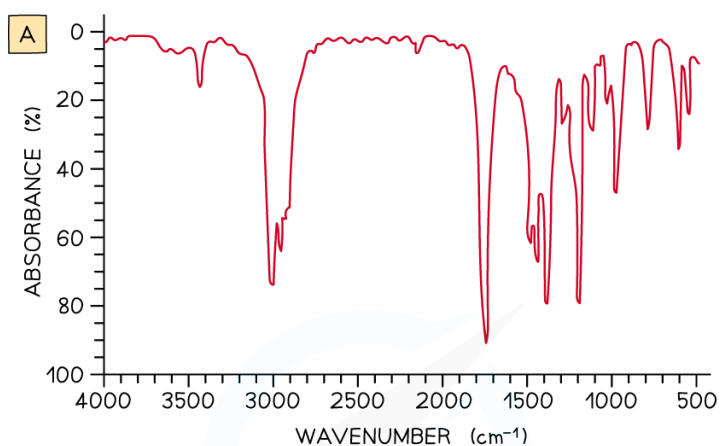
Worked Example

Analysing IR Spectra

Look at the two infrared spectra below and determine which one corresponds to propanone and which one to propan-2-ol



Your notes



Answer

- IR spectrum **A** is **propanone** and spectrum **B** is **propan-2-ol**.
 - In IR spectrum **A** the presence of a strong, sharp absorption around 1710 cm^{-1} corresponds to the characteristic $\text{C}=\text{O}$, carbonyl, group in a ketone.
 - In spectrum **B** the presence of a strong, broad absorption around $3200\text{--}3500\text{ cm}^{-1}$ suggests that there is an alcohol group present, which corresponds to the -OH group in propan-2-ol.